



Advanced Time Series Analysis and Forecasting

Chapter 4: Cointegration revisited: VECM, ARDL and panel data



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Acronyms used in this material (1/2)

- ▣ **ADF**: Augmented Dickey–Fuller (test)
- ▣ **AI**: Artificial Intelligence
- ▣ **AR**: AutoRegressive (model); in event studies: Abnormal Return
- ▣ **ARDL**: AutoRegressive Distributed Lag (model)
- ▣ **BIC**: Bayesian Information Criterion
- ▣ **BNR**: Banca Națională a României (the National Bank of Romania)
- ▣ **CADF**: Cross-sectionally augmented Dickey–Fuller (regression)
- ▣ **CCE**: Common Correlated Effects (estimator)
- ▣ **CCEMG**: Common Correlated Effects Mean Group
- ▣ **CD**: Cross-section Dependence (test of Pesaran)
- ▣ **CIPS**: Cross-sectionally augmented IPS (test)
- ▣ **CS-ARDL**: Cross-Sectionally augmented ARDL
- ▣ **DOLS**: Dynamic Ordinary Least Squares
- ▣ **ECM**: Error Correction Model
- ▣ **EU**: European Union
- ▣ **EU-27**: the 27 member states of the European Union
- ▣ **FE**: Fixed Effects
- ▣ **FEVD**: Forecast Error Variance Decomposition
- ▣ **FMOLS**: Fully Modified Ordinary Least Squares
- ▣ **FRED**: Federal Reserve Economic Data
- ▣ **GARCH**: Generalised AutoRegressive Conditional Heteroskedasticity

Acronyms used in this material (2/2)

- ▣ **GDP:** Gross Domestic Product
- ▣ **GMM:** Generalized Method of Moments
- ▣ **HICP:** Harmonised Index of Consumer Prices
- ▣ **IMF:** International Monetary Fund
- ▣ **IPS:** Im–Pesaran–Shin (panel unit-root test)
- ▣ **LLC:** Levin–Lin–Chu (panel unit-root test)
- ▣ **LLM:** Large Language Model
- ▣ **LR:** Likelihood Ratio (test)
- ▣ **MG:** Mean Group (estimator)
- ▣ **ML:** Machine Learning
- ▣ **NARDL:** Nonlinear AutoRegressive Distributed Lag (model)
- ▣ **NPISH:** Non-Profit Institutions Serving Households
- ▣ **OECD:** Organisation for Economic Co-operation and Development
- ▣ **OLS:** Ordinary Least Squares
- ▣ **PMG:** Pooled Mean Group (estimator)
- ▣ **PPP:** Purchasing Power Parity
- ▣ **PSS:** Pesaran–Shin–Smith
- ▣ **ROBOR:** Romanian Interbank Offer Rate
- ▣ **SE:** Standard Error
- ▣ **TSA:** Time Series Analysis
- ▣ **US:** United States
- ▣ **VAR:** Vector AutoRegression
- ▣ **VECM:** Vector Error Correction Model

Today's question and route

- **Question:** when do non-stationary series share a long-run equilibrium, which restrictions on it can we test, and how far can we trust the answer in samples of 100–300 observations or across 27 countries?
 - ▶ cointegration is a statement about the likelihood of a reduced-rank VAR; inference on it is non-standard and fragile
- **Route** of the chapter
 - ▶ the cointegrated VAR as a likelihood problem; the five deterministic cases; small-sample corrections and the bootstrap
 - ▶ identification and tests on β and α ; interest-rate pass-through in Romania; $I(2)$ in brief; common trends
 - ▶ ARDL and the bounds test; panel unit roots, panel cointegration and heterogeneous panel estimators
- We build on [TSA, Chapter 7](#) (Engle–Granger, ECM, the VECM, Johansen tests, pairs trading) and on [Chapter 3](#) (structural identification); Seminar 4 comes before this lecture

Learning outcomes

- ▣ Derive the Johansen estimator as a reduced-rank regression, state the asymptotic distribution of the trace statistic and choose the deterministic case
- ▣ Correct the rank test in small samples (Reinsel–Ahn, Bartlett, wild bootstrap) and report the sensitivity of the rank to the lag length
- ▣ Identify cointegrating vectors, test economic restrictions on β and weak exogeneity through α , and build a common-trends structural VECM
- ▣ Run the ARDL bounds test with correct (and small-sample) critical values, and decide between ARDL and VECM
- ▣ Test for cross-section dependence, panel unit roots and panel cointegration, and estimate long-run panel relations by MG, PMG and CCE

Reading, data and tools

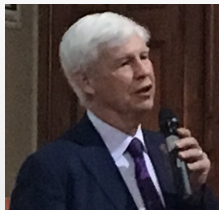
- ▣ Backbone: [Johansen \(1995a\)](#); [Juselius \(2006\)](#); [Kilian and Lütkepohl \(2017\)](#), Ch. 3 and 10; [Pesaran \(2015b\)](#), Ch. 22–31; [Lütkepohl \(2005\)](#), Ch. 6–9
 - ▶ original papers: [Johansen \(1988\)](#), [Johansen \(1991\)](#), [Pesaran, Shin and Smith \(2001\)](#), [Pesaran, Shin and Smith \(1999\)](#), [Pesaran \(2006\)](#), [Pesaran \(2007\)](#); survey: [Breitung and Pesaran \(2008\)](#)
- ▣ Python Quantlets of this chapter: [Quantlets/Ch_04](#)
 - ▶ Johansen in five cases, restricted ML, bootstrap rank tests, ARDL bounds, LLC/IPS/CIPS, Pedroni, Westerlund, MG, PMG and CCE written out in `numpy`
- ▣ Lecture notebook: [open in Google Colab](#)

Data used in this chapter

Series	Source	Use
Romanian lending and deposit rates (lei); ROBOR 3M	IMF International Financial Statistics (BNR data); Eurostat irt_st_m	VECM, ARDL
US consumption of nondurables and services, fixed investment, GDP, deflator, population	FRED (PCND, PCESV, FPI, GDP, GDPDEF, B230RC0Q173SBEA)	common trends
Romanian HICP, monthly	Eurostat prc_hicp_minr	I(2)
EU-27 household consumption, disposable income, consumption deflator, population	Eurostat nama_10_gdp, nasa_10_nf_tr, nama_10_pe	panel

- All sources are public and need no account or key; monthly rates to March 2026, US data to 2025, EU annual data to 2024
- HICP: harmonised index of consumer prices; NPISH: non-profit institutions serving households (included with households)

From spurious regressions to a likelihood theory



Robert Engle, 2017

Photo: [Econterms \(2017\)](#); CC BY-SA 4.0; [Wikimedia Commons](#)

- 1987: [Engle and Granger \(1987\)](#) define cointegration and prove the representation theorem; estimation is a two-step regression
- 1988–1991: [Johansen \(1988\)](#) and [Johansen \(1991\)](#) turn it into maximum likelihood for a VAR with reduced rank: several relations, tests on their coefficients
- 1999–2001: Pesaran, Shin and Smith add ARDL bounds tests [Pesaran, Shin and Smith \(2001\)](#) and pooled panels [Pesaran, Shin and Smith \(1999\)](#)
- 2003: Engle and Granger share the Nobel Prize; Granger “for methods of analyzing economic time series with common trends (cointegration)”
- Since then: bootstrap rank tests, cross-section dependence in panels, common factors

From TSA to this chapter

- ▣ Known (TSA, Chapter 7): $y_t \sim I(1)$, $\beta' y_t \sim I(0)$; Engle–Granger two-step; the ECM; $\Pi = \alpha\beta'$; trace and max-eigenvalue tests with tabulated critical values
- ▣ New here
 - ▶ why the eigenvalue problem is the maximum likelihood estimator, and what its limit distribution looks like
 - ▶ which deterministic case to use, and what goes wrong in samples of 100–300 months
 - ▶ restrictions on β (economic hypotheses) and on α (weak exogeneity); structural shocks with permanent effects
 - ▶ single equations (ARDL) and panels of countries when the system approach is too demanding

The VECM and the hypothesis $H(r)$ (1/2)

- The VECM: this month's change of each variable responds to last month's levels, to past changes and to deterministic terms

$$\Delta y_t = \Pi y_{t-1} + \sum_{i=1}^{p-1} \Gamma_i \Delta y_{t-i} + \Phi D_t + \varepsilon_t$$

- Notation

- ▶ $y_t \in \mathbb{R}^n$: the vector of the n $I(1)$ variables; $\Delta y_t = y_t - y_{t-1}$: their changes
- ▶ Π ($n \times n$): the long-run (levels) matrix; Γ_i ($n \times n$): the short-run matrices; p : the lag order of the VAR in levels
- ▶ D_t : deterministic terms (constant, trend, dummies) with coefficients Φ
- ▶ $\varepsilon_t \sim \text{i.i.d. } N(0, \Omega)$: the innovations, with covariance matrix Ω

- Gaussian likelihood conditional on the first p observations: the estimator is ML, the tests are likelihood ratios Johansen (1988), Johansen (1991)

The VECM and the hypothesis $H(r)$ (2/2)

- $H(r)$: Π has rank at most r , so it factors as $\Pi = \alpha\beta'$ with α, β of dimension $n \times r$
 - ▶ r : the number of cointegrating relations, $0 \leq r \leq n$
 - ▶ β : its columns are the r cointegrating vectors; $\beta'y_t$ is stationary (the long run)
 - ▶ α : the adjustment (loading) coefficients; α_{ij} is the response of $\Delta y_{i,t}$ to the deviation $\beta'_j y_{t-1}$
- The models are nested: $H(0) \subset H(1) \subset \dots \subset H(n)$
 - ▶ $H(n)$: unrestricted VAR in levels; $H(0)$: VAR in differences
- Only Π is identified, not α and β separately: $\alpha\beta' = (\alpha\xi)(\beta\xi^{-1'})'$
 - ▶ ξ : any invertible $r \times r$ matrix; $\xi^{-1'}$: the transpose of its inverse

Step 1: concentrating out the short run

- Regress Δy_t and y_{t-1} on $Z_{2t} = (\Delta y'_{t-1}, \dots, \Delta y'_{t-p+1}, D'_t)'$ and keep the residuals (Frisch–Waugh)
 - ▶ Z_{2t} : all short-run regressors (lagged changes and deterministic terms)
 - ▶ R_{0t}, R_{1t} : the residuals of Δy_t and of y_{t-1} , i.e. both purged of the short run
- The concentrated model is a regression with a reduced-rank coefficient: $R_{0t} = \alpha \beta' R_{1t} + \hat{\varepsilon}_t$
- Moment matrices $S_{ij} = T^{-1} \sum_t R_{it} R'_{jt}$, $i, j \in \{0, 1\}$
 - ▶ T : the number of observations used; S_{00}, S_{11} : the covariance matrices of R_{0t}, R_{1t} ; $S_{01} = S'_{10}$: their cross-covariance
- For fixed β , OLS of R_{0t} on $\beta' R_{1t}$ gives α and Ω in closed form:
 - ▶ $\hat{\alpha}(\beta) = S_{01} \beta (\beta' S_{11} \beta)^{-1}$, $\hat{\Omega}(\beta) = S_{00} - S_{01} \beta (\beta' S_{11} \beta)^{-1} \beta' S_{10}$
 - ▶ $L_{\max}(\beta)$: the likelihood maximised over all parameters except β ; $L_{\max}^{-2/T}(\beta) \propto |\hat{\Omega}(\beta)|$ ($|\cdot|$: determinant)
 - ▶ what remains is a problem in β alone: minimise $|\hat{\Omega}(\beta)|$

Step 2: the eigenvalue problem

- Determinant identity: $|\hat{\Omega}(\beta)| = |S_{00}| \frac{|\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|}{|\beta'S_{11}\beta|}$
- Minimising this ratio of quadratic forms is the generalised eigenvalue problem $|\lambda S_{11} - S_{10}S_{00}^{-1}S_{01}| = 0$
 - its n roots, ordered, are the eigenvalues $1 > \hat{\lambda}_1 > \dots > \hat{\lambda}_n > 0$; $\hat{v}_i$: the eigenvector of $\hat{\lambda}_i$
- $\hat{\beta}$ = the eigenvectors $\hat{v}_1, \dots, \hat{v}_r$ of the r largest eigenvalues; then $L_{\max}^{-2/T} = |S_{00}| \prod_{i=1}^r (1 - \hat{\lambda}_i)$
 - normalisation $\hat{\beta}' S_{11} \hat{\beta} = I_r$ (I_r : the $r \times r$ identity matrix)
- $\hat{\lambda}_i$ are the squared canonical correlations between R_{0t} and R_{1t} : large $\hat{\lambda}_i$ = a linear combination of levels that predicts the changes
- One eigen-decomposition gives the ML estimates for every r at once (proof: [► Appendix](#))

Likelihood ratio tests of the rank (1/2)

- **Trace:** $H(r)$ against $H(n)$: $LR_{tr}(r) = -T \sum_{i=r+1}^n \ln(1 - \hat{\lambda}_i)$
 - ▶ sums the $n - r$ smallest eigenvalues, those that $H(r)$ sets to zero
 - ▶ $\hat{\lambda}_i \approx 0$ gives $-\ln(1 - \hat{\lambda}_i) \approx 0$: a small statistic supports $H(r)$; large values reject it
- **Maximum eigenvalue:** $H(r)$ against $H(r + 1)$: $LR_{\max}(r) = -T \ln(1 - \hat{\lambda}_{r+1})$
 - ▶ uses only the next eigenvalue, $\hat{\lambda}_{r+1}$: is there one more relation?
- **Rank determination:** test $r = 0, 1, \dots$ in turn and stop at the first non-rejection; this sequence is consistent for the true rank at a fixed level [Johansen \(1995a\)](#)

Likelihood ratio tests of the rank (2/2)

- Limit under $H(r)$:

$$LR_{tr} \Rightarrow \text{tr} \left\{ \int_0^1 (dB) F' \left(\int_0^1 FF' du \right)^{-1} \int_0^1 F (dB)' \right\}$$

- Notation

- ▶ $\Rightarrow$: convergence in distribution as $T \rightarrow \infty$; tr : the trace (sum of the diagonal elements)
- ▶ $B(u)$, $u \in [0, 1]$: an $(n - r)$ -dimensional standard Brownian motion, the limit of the rescaled common trends
- ▶ $F = B$ corrected for the deterministic terms of the case

- The limit depends only on $n - r$ and the case, not on $\alpha, \beta, \Gamma_i, \Omega$

- ▶ a multivariate Dickey–Fuller distribution: quantiles by simulation [Osterwald-Lenum \(1992\)](#), [MacKinnon, Haug and Michelis \(1999\)](#)
- ▶ non-standard: neither χ^2 nor Normal

The Granger representation theorem (1/3)



Clive Granger, 2008

Photo: Olaf Storbeck (2008); CC BY-SA 2.0;

Wikimedia Commons

- The theorem writes a cointegrated VAR as random-walk trends plus stationary deviations
- VAR in levels: $y_t = \sum_{i=1}^p A_i y_{t-i} + \Phi D_t + \varepsilon_t$
 - ▶ A_i ($n \times n$): its lag matrices; $A(z) = I - \sum_{i=1}^p A_i z^i$: its characteristic polynomial
 - ▶ $\Pi = -A(1)$, $\Gamma_i = -\sum_{j>i} A_j$: the VECM matrices; $\Gamma = I - \sum_i \Gamma_i$
- Conditions
 - ▶ the roots of $|A(z)| = 0$ satisfy $|z| > 1$ or $z = 1$ (no explosive or seasonal roots)
 - ▶ $\text{rank } \Pi = r$, so $\Pi = \alpha\beta'$
 - ▶ $\alpha'_\perp \Gamma \beta_\perp$ has full rank $n - r$
 - ▶ $\alpha_\perp, \beta_\perp$ ($n \times (n - r)$): orthogonal complements, $\alpha'_\perp \alpha_\perp = 0$, $\beta'_\perp \beta_\perp = 0$

The Granger representation theorem (2/3)

- Under these conditions y_t is $I(1)$, $\beta'y_t$ is $I(0)$, and

$$y_t = C \sum_{i=1}^t (\varepsilon_i + \Phi D_i) + C^*(L)(\varepsilon_t + \Phi D_t) + A_0$$

- Notation

- ▶ $C = \beta_{\perp}(\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp}$: the long-run impact matrix, of rank $n - r$ (derivation: [Appendix](#))
- ▶ $C^*(L) = \sum_{j \geq 0} C_j^* L^j$: a polynomial in the lag operator L ($Ly_t = y_{t-1}$) with summable coefficients, the stationary part
- ▶ A_0 : a constant that depends on the initial values, with $\beta' A_0 = 0$

- First term: random walks plus deterministic trends; second term: stationary deviations from them

The Granger representation theorem (3/3)

□ Interpretation

- ▶ $\beta' C = 0$: the relations do not contain the stochastic trends
- ▶ $C\alpha = 0$: an equilibrium error has no permanent effect
- ▶ common trends: the $n - r$ random walks $\alpha'_{\perp} \sum_i \varepsilon_i$

□ If $\alpha'_{\perp} \Gamma \beta_{\perp}$ is singular, the theorem fails

- ▶ C does not exist and some trends are $I(2)$ (see the $I(2)$ section)

Recap: The cointegrated VAR

- ▣ Concentrate out the short run, then maximise over β : a generalised eigenvalue problem
- ▣ The eigenvalues are squared canonical correlations; the trace and max-eigenvalue statistics are likelihood ratios
- ▣ Their limits are Brownian functionals that depend on $n - r$ and on the deterministic terms only
- ▣ Granger representation: $C = \beta_{\perp}(\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp}$ links α, β to the common trends

Five ways to place a constant and a trend

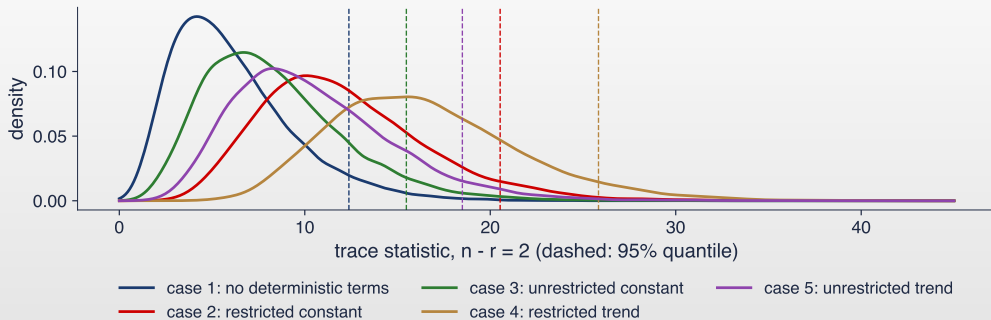
- $\mu_t = \mu_0 + \mu_1 t$ with $\mu_j = \alpha \rho_j + \alpha_{\perp} \gamma_j$: the α part sits in the relations, the $\alpha_{\perp}$ part drives the levels [Johansen \(1995a\)](#)

► μ_t : constant μ_0 plus trend $\mu_1 t$ in the VECM; ρ_j ($r \times 1$), γ_j ($((n-r) \times 1)$): their coordinates in the two directions

	Case	Restriction	Levels y_t	Relations $\beta' y_t$
	1 (H_2)	$\mu_0 = \mu_1 = 0$	no drift	zero mean
□	2 (H_1^*)	$\mu_0 = \alpha \rho_0, \mu_1 = 0$	no drift	constant
	3 (H_1)	$\mu_1 = 0$	linear trend	constant
	4 (H^*)	$\mu_1 = \alpha \rho_1$	linear trend	linear trend
	5 (H)	none	quadratic trend	linear trend

- “Restricted” terms enter Z_{1t} (the regressors multiplied by Π) with y_{t-1} , as part of β ; “unrestricted” terms enter Z_{2t} and are concentrated out
- Interest rates: case 2; trending macro aggregates: case 3 or 4; case 5 is rarely plausible

The case changes the null distribution



- Trace statistic under $H(r)$ with $n - r = 2$, simulated from random walks (10 000 replications, $T = 400$; drift in cases 3 and 5)
- 95% quantiles for every case and $n - r = 1, \dots, 4$: the table two slides later

Interpreting the five distributions

- Each restricted deterministic term adds a regressor to the reduced-rank problem and shifts the distribution to the right
- A trend in the levels (cases 3 and 5) makes one direction of the common trend deterministic: for $n - r = 1$ the limit is $\chi^2(1)$
- Using case-3 critical values in a case-2 model over-rejects: the 95% quantile for $n - r = 1$ is 9.25 in case 2 and 3.89 in case 3

Simulated 95% quantiles of the trace statistic

$n - r$	case 1	case 2	case 3	case 4	case 5
1	4.21	9.25	3.89	12.58	3.84
2	12.38	20.53	15.48	25.84	18.50
3	24.44	35.24	29.78	43.14	35.36
4	40.29	53.68	47.98	63.85	55.54

- Simulated with 10 000 replications of a VAR(1) test regression on $T = 400$ observations; the response-surface critical values of [MacKinnon, Haug and Michelis \(1999\)](#) differ by at most a few tenths
- Choosing the case: plot the data first; if unsure between 2 and 3 (or 3 and 4), test the joint hypothesis on rank and case in the order of the Pantula principle [Pantula \(1989\)](#), [Johansen \(1992\)](#)
- Breaks in the deterministic terms shift the distribution again: use the critical values of [Johansen, Mosconi and Nielsen \(2000\)](#) for broken trends

Recap: Deterministic terms

- Split each deterministic term into an α part (in the relations) and an $\alpha_{\perp}$ part (in the levels)
- The five cases have five different null distributions; mixing them up changes the rank
- Plot the data, choose the case on economic grounds and use the Pantula principle when in doubt

Over-rejection of the asymptotic test (1/2)

- The VECM spends many parameters on the short run
 - ▶ $n^2(p-1)$ elements of $\Gamma_1, \dots, \Gamma_{p-1}$, plus the deterministic terms
 - ▶ with $T = 100$, $n = 3$ and $p = 4$: 27 short-run parameters, a large share of the information
- The distortion grows with persistence
 - ▶ persistent short-run dynamics: roots of the stationary part close to the unit circle
 - ▶ the asymptotic test then finds too many cointegrating relations
- Conditional heteroskedasticity [Cavaliere, Rahbek and Taylor \(2010\)](#)
 - ▶ leaves the limit distribution unchanged
 - ▶ but worsens the finite-sample size: use the wild bootstrap

Over-rejection of the asymptotic test (2/2)

▣ **Reinsel–Ahn** Reinsel and Ahn (1992)

- ▶ multiply the statistic by $(T - np)/T$: a degrees-of-freedom correction
- ▶ simple, often too conservative

▣ **Bartlett correction** Johansen (2002)

- ▶ $\mathbb{E}LR_{tr} \approx f(1 + a(\theta)/T)$; the corrected statistic is $LR_{tr}/(1 + a(\theta)/T)$
- ▶ f : the mean of the limit distribution; θ : the VECM parameters
- ▶ $a(\theta)$: an analytic function of θ , evaluated at the estimates

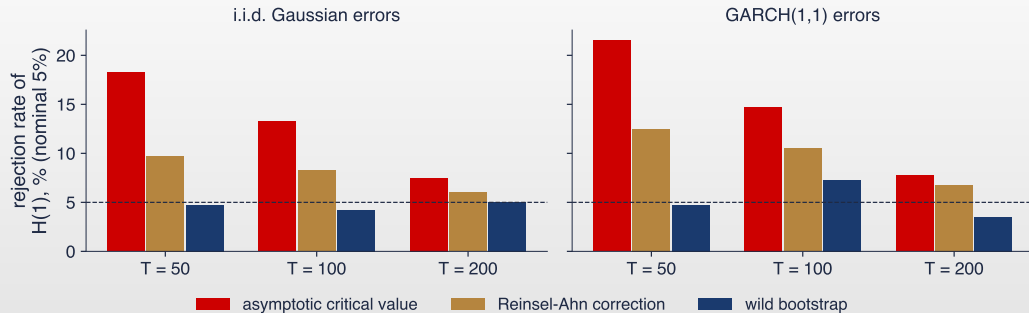
▣ **Bootstrap** Swensen (2006), Cavaliere, Rahbek and Taylor (2012)

- ▶ simulate the null distribution from the model estimated under $H(r)$

The bootstrap rank test, step by step

- ▣ 1. Estimate the VECM under $H(r)$: $\hat{\alpha}^{(r)}, \hat{\beta}^{(r)}, \hat{\Gamma}_i^{(r)}, \hat{\Phi}^{(r)}$ and residuals $\hat{\varepsilon}_t^{(r)}$
- ▣ 2. Generate $\Delta y_t^* = \hat{\alpha}^{(r)} \hat{\beta}^{(r)'} y_{t-1}^* + \sum_i \hat{\Gamma}_i^{(r)} \Delta y_{t-i}^* + \hat{\Phi}^{(r)} D_t + \varepsilon_t^*$ from the observed initial values
- ▣ 3. Innovations: $\varepsilon_t^* = \hat{\varepsilon}_t^{(r)} w_t$, $w_t \sim \text{i.i.d. } N(0, 1)$ (wild) or resampled residuals (i.i.d. bootstrap)
 - the star marks bootstrap quantities; w_t : a random sign-and-scale weight; the wild version keeps the volatility pattern of each date
- ▣ 4. Compute $LR_{tr}^*(r)$ on each sample; p-value = share of $LR_{tr}^* \geq LR_{tr}(r)$; test $r = 0, 1, \dots$ and stop at the first non-rejection
- ▣ Estimating under $H(r)$ (not under $H(n)$) is essential: the bootstrap data must have exactly $n - r$ unit roots
Cavaliere, Rahbek and Taylor (2012)

Size of the rank test in small samples (simulation)



- Three-variable VECM, true rank 1, $\Gamma_1 = 0.8I$, case 2, VAR(2); test of $H(1)$ at 5%; 400 replications, 199 bootstrap samples each
- $T = 50$, i.i.d. errors: asymptotic 18.2%, Reinsel-Ahn 9.8%, wild bootstrap 4.8%

Interpreting the size simulation

- ▣ With persistent short-run dynamics the asymptotic test finds a spurious second relation in 18.2% of samples of 50 and 13.2% of samples of 100
- ▣ Reinsel–Ahn halves the distortion; the wild bootstrap is close to 5% at every T , also under GARCH errors (4.8% at $T = 50$, asymptotic 21.5%)
- ▣ At $T = 200$ all three are acceptable: the problem is the ratio of parameters to observations, not the method
- ▣ Practice: report asymptotic and bootstrap p -values side by side, and the rank for neighbouring lag lengths

Recap: Small samples

- The asymptotic trace test over-rejects when parameters are many and the short run is persistent
- Reinsel–Ahn and the Bartlett correction rescale the statistic; the bootstrap rebuilds its distribution
- Bootstrap from the model estimated under $H(r)$; use wild weights when volatility changes over time

Normalisation is not identification (1/2)

- The eigenvectors give one basis of the cointegration space $\text{sp}(\beta)$; any $\beta\xi$ spans the same space and fits equally well
 - ▶ $\text{sp}(\beta)$: the space spanned by the columns of β ; ξ : an invertible $r \times r$ matrix (a rotation)
- Dividing each vector by one coefficient fixes the scale (r restrictions), but not the rotation
 - ▶ $r(r - 1)$ further restrictions are needed, $r - 1$ per vector
- Identification rests on economic theory: each restriction is a statement such as “the lending rate does not depend on the deposit rate in the long run”

Normalisation is not identification (2/2)

- Linear restrictions vector by vector: $\beta = (H_1\varphi_1, \dots, H_r\varphi_r)$
 - ▶ n_1 : the length of each vector (n plus the restricted deterministic terms)
 - ▶ H_i ($n_1 \times s_i$, known): the design matrix of vector i ; φ_i ($s_i \times 1$): its s_i free coefficients
 - ▶ $R_i = H_{i\perp}$: the restrictions written as $R_i'\beta_i = 0$
- Rank condition [Johansen \(1995b\)](#): for every i and every set of k other vectors $i_1, \dots, i_k$
 - ▶ $\text{rank}(R_i'(H_{i_1}, \dots, H_{i_k})) \geq k$, $k = 1, \dots, r-1$
 - ▶ meaning: no combination of other vectors satisfies the restrictions of vector i
 - ▶ the cointegration analogue of the rank condition in simultaneous equations; the order condition alone is not enough

Testing restrictions on β (1/2)

- **The same restriction on all vectors**, $\beta = H\varphi$: solve $|\lambda H' S_{11} H - H' S_{10} S_{00}^{-1} S_{01} H| = 0$
 - ▶ H ($n_1 \times s$, known): the restriction; φ ($s \times r$): the free coefficients; e.g. excluding one variable from all relations
 - ▶ $\tilde{\lambda}_i$: the eigenvalues of the restricted problem; $\hat{\lambda}_i$: the unrestricted ones
- Test statistic: $LR = T \sum_{i=1}^r \ln\{(1 - \tilde{\lambda}_i)/(1 - \hat{\lambda}_i)\} \rightarrow \chi^2(r(n_1 - s))$
 - ▶ $LR \geq 0$; it is large when the restriction lowers the canonical correlations, i.e. when the data reject it
 - ▶ degrees of freedom $r(n_1 - s)$: $n_1 - s$ restrictions on each of the r vectors
- Given the rank, $\hat{\beta}$ is super-consistent and mixed Gaussian: LR tests on β are χ^2 , unlike the rank test [Johansen \(1991\)](#)

Testing restrictions on β (2/2)

- ▣ **Different restrictions per vector**, $\beta = (H_1\varphi_1, \dots, H_r\varphi_r)$: no closed form
 - ▶ maximise the concentrated log-likelihood $-\frac{T}{2} \ln |\hat{\Omega}(\beta)|$ by the switching algorithm (one vector at a time) or numerically
 - ▶ over-identifying restrictions: $LR \rightarrow \chi^2(\sum_i (n_1 - r + 1 - s_i))$; each vector has $n_1 - r + 1 - s_i$ restrictions beyond the $r - 1$ needed to identify it
- ▣ **Fully known** $\beta = \beta_0$ (e.g. the great ratios):

$$LR = T \left\{ \ln |\hat{\Omega}(\beta_0)| - \ln |S_{00}| - \sum_{i \leq r} \ln(1 - \hat{\lambda}_i) \right\} \rightarrow \chi^2(r(n_1 - r))$$

- ▶ compares the fit with β_0 imposed ($\hat{\Omega}(\beta_0)$) to the unrestricted fit; β_0 : the hypothesised matrix

Restrictions on α and weak exogeneity

- $\alpha = A\psi$: only m combinations of the equations adjust to the relations
 - ▶ A ($n \times m$, known): the restriction; ψ ($m \times r$): the free adjustment coefficients; $A_{\perp}$: its orthogonal complement
 - ▶ condition the system on $A'_{\perp} R_{0t}$ and solve the eigenvalue problem of the remaining m equations (eigenvalues $\tilde{\lambda}_i$)
 - ▶ $LR = T \sum_{i \leq r} \ln\{(1 - \tilde{\lambda}_i)/(1 - \hat{\lambda}_i)\} \rightarrow \chi^2(r(n - m))$
- **Weak exogeneity** of x_t for β : the row of α for x_t is zero (a zero row of α , r restrictions)
 - ▶ x_t does not adjust to past disequilibria; it is a common trend (a column of $\alpha_{\perp}$)
 - ▶ the conditional model of Δy_t given Δx_t is then fully efficient for β : the basis of single-equation ECM and ARDL
- Long-run structural hypotheses are joint restrictions on β and α
 - ▶ PPP: $(1, -1, -1)$ on (price, foreign price, exchange rate); Fisher: $(1, -1)$ on (interest rate, inflation); complete pass-through: $(1, -1)$ on (retail rate, policy or market rate)

Recap: Identification and tests

- ▣ Identify β with $r - 1$ economically motivated restrictions per vector; check the rank condition
- ▣ Given the rank, LR tests on β and α are χ^2 with known degrees of freedom
- ▣ A zero row of α = weak exogeneity = permission to condition on that variable

How fast and how fully do banks follow ROBOR?



The BNR palace, Bucharest

Photo: Ștefan Jurcă (2015); CC BY 2.0; Wikimedia

[Commons](#)

- Monetary policy reaches households and firms only through bank rates; ROBOR 3M is the reference for most variable-rate loans in lei
- Euro area: long-run pass-through to lending rates often complete, to deposit rates incomplete [de Bondt \(2005\)](#); CEE: faster and fuller pass-through after the early transition [Égert, Crespo-Cuaresma and Reininger \(2007\)](#)
- Hypotheses: H_β : lending and deposit rates move one for one with ROBOR in the long run; H_α : ROBOR does not adjust to bank rates
- Data: August 2005 – March 2026, $T = 248$ months (inflation targeting)

Lending rate, deposit rate and ROBOR 3M



- Monthly, % per year; lending and deposit rates in lei (IMF IFS, reported by the BNR); ROBOR 3M (Eurostat)
- ROBOR ranges from 0.51% to 18.2%; the lending–ROBOR spread falls from 11.3 pp to 2.5 pp

Interpreting the three rates

- All three rates wander far from any mean over twenty years: treat them as $I(1)$ within the sample, even if interest rates are bounded in theory
- They move together after 2008, in 2014–2015 and in 2022: the visual case for cointegration
- The falling spread in 2005–2010 is the end of the transition (competition, falling risk premia), not a policy effect
- No trend in levels is plausible for rates: case 2, constant restricted to the relations

Rank of the pass-through system

$H(r)$	LR_{tr}	RA	cv 5%	wild p	$LR_{tr}, p=4$	RA, $p=4$	wild $p, p=4$
$r = 0$	121.1	118.1	35.2	0.007	40.0	38.0	0.647
$r \leq 1$	52.2	50.9	20.5	0.022	14.9	14.1	0.812
$r \leq 2$	4.3	4.2	9.3	0.767	3.8	3.6	0.820

- Case 2; RA: Reinsel–Ahn; cv: simulated 95% quantile; wild p : 399 wild-bootstrap samples under $H(r)$; lag order by BIC = 2, HQ = 5, AIC = 8
- With $p = 2$ all three methods give $r = 2$; with $p = 4$ the asymptotic test gives $r = 1$ and the wild bootstrap rejects nothing

Interpreting the rank tests

- ▣ Two relations are what theory predicts: one for lending rates, one for deposit rates, both anchored to ROBOR
- ▣ The result is fragile: two extra lags and heteroskedasticity-robust inference remove the evidence; the 2008–2009 volatility dominates the sample
- ▣ We continue with $r = 2$ because theory and the parsimonious model agree, and we report the sensitivity
- ▣ A rank chosen this way is a modelling decision, not a measurement: state it before testing restrictions

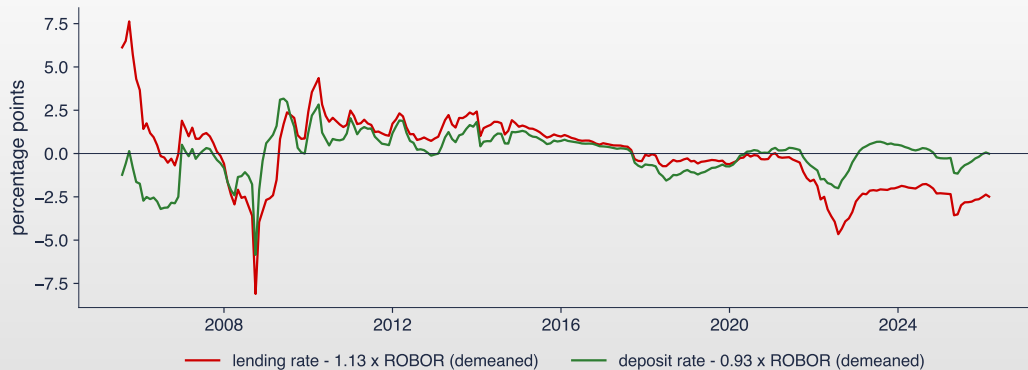
Identified relations and the tests

- Just-identified (one exclusion per vector): $\text{lend}_t = 1.13 \text{ROBOR}_t + 3.62$, $\text{dep}_t = 0.93 \text{ROBOR}_t + 0.09$
- Adjustment: lending rate $\alpha_{11} = -0.056$ (to its own relation), deposit rate $\alpha_{22} = -0.071$; ROBOR: 0.014 and 0.017

	Hypothesis	LR	df	p
□	complete pass-through to both rates	8.98	2	0.011
	complete pass-through to the lending rate	4.16	1	0.041
	ROBOR weakly exogenous (zero row of α)	0.57	2	0.752

- With $p = 4$: lending pass-through 1.11, p -value of complete pass-through to lending 0.458, of weak exogeneity 0.819

The two equilibrium errors



□ $\hat{\beta}'_i y_t$ for the just-identified relations, demeaned; they should look stationary if the rank and β are right

Interpreting the pass-through VECM

- Lending rates over-react slightly in the long run (1.13 per point of ROBOR), deposit rates under-react (0.93): the bank margin widens when ROBOR rises
- Joint complete pass-through is rejected ($p = 0.011$), driven by deposits; for lending rates alone the verdict depends on the lag length
- ROBOR is weakly exogenous ($p = 0.752$): money-market rates lead, bank rates follow; this justifies a single-equation ARDL
- The lending equilibrium error is persistent before 2010: the transition-era spread is not a stable markup, a candidate for a broken constant [Johansen, Mosconi and Nielsen \(2000\)](#)

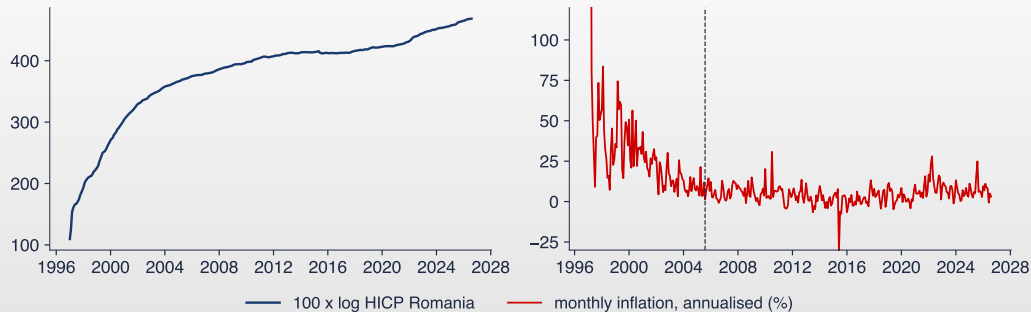
Recap: Pass-through in Romania

- Two cointegrating relations with ROBOR as the common trend, but the rank is sensitive to lags and to heteroskedasticity
- Lending pass-through close to (slightly above) one, deposit pass-through below one
- Weak exogeneity of ROBOR is not rejected: a conditional single-equation model is legitimate

When the I(1) model is not enough

- If $\alpha'_{\perp} \Gamma \beta_{\perp}$ has reduced rank $s < n - r$, the Granger representation fails
 - ▶ some common trends are I(2): y_t contains double sums $\sum_{j \leq t} \sum_{i \leq j} \varepsilon_i$
 - ▶ typical candidates: nominal prices and nominal money when inflation changes; inflation is then I(1)
- The I(2) model [Johansen \(1995c\)](#), [Juselius \(2006\)](#)
 - ▶ two reduced-rank conditions: $\Pi = \alpha \beta'$ and $\alpha'_{\perp} \Gamma \beta_{\perp} = \xi \eta'$
 - ▶ $\xi, \eta ((n - r) \times s, \text{full rank})$: the factors of the second reduced-rank matrix
 - ▶ polynomial cointegration: $\beta' y_t + \delta' \Delta y_t \sim I(0)$; $\delta (n \times r)$: the weights of the differences
 - ▶ example: real money plus a multiple of inflation
- Practical check
 - ▶ test Δy_t for a unit root; a root near one in the stationary part of the I(1) VECM is a warning
 - ▶ the usual fix: model real (deflated) variables and inflation

Is the Romanian price level I(2)?



- Left: 100 ln HICP; right: monthly inflation, annualised; dashed line: inflation targeting from August 2005
- ADF t on monthly inflation (12 lags): -4.25 for 1997–2026, -2.39 since 2005 (5% critical value -2.87); on its change: -6.13

Interpreting the I(2) check

- ▣ Over 1997–2026, disinflation looks like mean reversion and inflation tests as stationary: the price level is I(1)
- ▣ Since 2005 a unit root in inflation is not rejected: within the inflation-targeting sample the price level behaves like I(2)
- ▣ The answer depends on the sample, as for the rank: this is why systems with nominal levels are usually rewritten in real terms and inflation
- ▣ Our pass-through system uses rates, which are at most I(1): the I(2) issue does not arise there

Permanent and transitory shocks (1/2)

- Granger representation: the long-run effect of ε_t on y_{t+h} , $h \rightarrow \infty$, is $C\varepsilon_t$, and $\text{rank } C = n - r$
- Structural shocks: $\varepsilon_t = Bu_t$, $\mathbb{E}u_t u_t' = I$
 - ▶ u_t : n uncorrelated shocks with unit variance; B ($n \times n$): their impact effects, $BB' = \Omega$
 - ▶ the long-run impact CB has at most $n - r$ non-zero columns: $k = n - r$ permanent and r transitory shocks
- Counting (Chapter 3 needed $n(n - 1)/2$ restrictions): cointegration gives rk zero long-run restrictions for free
 - ▶ still needed: $k(k - 1)/2$ among the permanent shocks and $r(r - 1)/2$ among the transitory shocks Kilian and Lütkepohl (2017), Gonzalo and Ng (2001)

Permanent and transitory shocks (2/2)

- With one common trend ($k = 1$) the permanent shock is identified without further assumptions
 - ▶ the shock: $u_t^P = \alpha'_\perp \varepsilon_t / (\alpha'_\perp \Omega \alpha_\perp)^{1/2}$, the innovation of the common trend scaled to unit variance
 - ▶ its impact: $b^P = \Omega \alpha_\perp (\alpha'_\perp \Omega \alpha_\perp)^{-1/2}$, the column of B that moves the n variables on impact
- This generalises Blanchard–Quah ([Chapter 3](#)): there the long-run zero was imposed; here it comes from the cointegration rank

Case study: King, Plosser, Stock and Watson (1991) (1/2)

□ King, Plosser, Stock and Watson (1991)

- ▶ a real business-cycle model with a stochastic productivity trend
- ▶ consumption, investment and output share one common trend
- ▶ the great ratios $c - y$ and $i - y$ are stationary (c, i, y : logarithms per capita)

□ In VECM terms

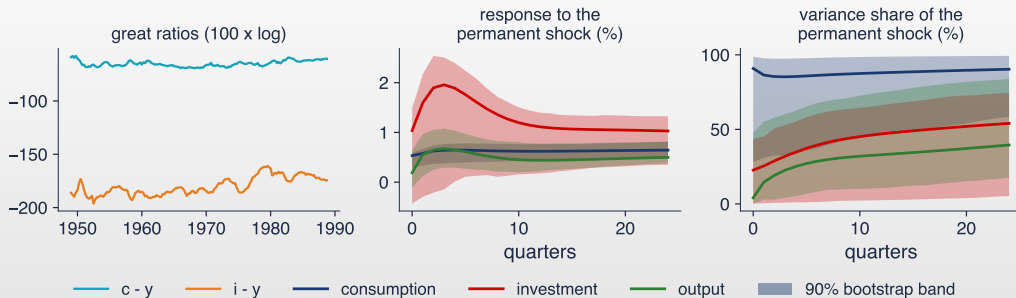
- ▶ $n = 3, r = 2, \beta = \begin{pmatrix} 1 & 0 \\ 0 & 1 \\ -1 & -1 \end{pmatrix}$ imposed for (c, i, y)
- ▶ $k = n - r = 1$: one permanent “balanced-growth” shock

□ Question: how much of the business cycle does the permanent shock explain?

Case study: King, Plosser, Stock and Watson (1991) (2/2)

- ▣ Our replication: US quarterly data, 1949Q1–1988Q4 (the paper's sample), $T = 160$
 - ▶ c : nondurables and services; i : fixed investment; y : GDP
 - ▶ all per capita, deflated by the GDP deflator
- ▣ Rank
 - ▶ VAR(2) in levels (all criteria), case 3
 - ▶ trace: 45.8, 19.4, 1.4 against 95% quantiles 29.8, 15.5, 3.9: $r = 2$
- ▣ Test of $\beta = \beta_0$ (the exact great ratios)
 - ▶ case 3: rejected, $LR = 16.6$, $\chi^2(2)$, $p < 0.001$
 - ▶ with a restricted trend (case 4): not rejected, $LR = 0.50$, $p = 0.778$

The balanced-growth shock



- Great ratios imposed (case 3); one-standard-deviation permanent shock; 90% residual-bootstrap bands (300 samples, β fixed)
- Long-run effect 0.74% on all three; share in output variance: 4% at 1 quarter, 30% at 8, 39% at 24 [17; 83]

Interpreting the common-trends model

- Balanced growth holds by construction in the long run: the shock moves c , i and y by the same 0.74%; investment overshoots in the short run
- The permanent shock dominates consumption (87% at 8 quarters, the permanent-income logic) but explains a minority of output fluctuations at business-cycle horizons
- Bands are wide: with $T = 160$ the data barely separate permanent from transitory variation
- As in [Chapter 3](#), the economics is in the identifying assumption: here, which relations are stationary

How robust is the answer?

Specification	$h = 1$	$h = 4$	$h = 8$	$h = 24$
great ratios, case 3, 1949–1988	4%	23%	30%	39%
great ratios with trend, case 4, 1949–1988	34%	37%	52%	77%
great ratios, case 3, 1949–2019	51%	30%	29%	40%
great ratios, case 3, 1949–2025	28%	20%	20%	29%

- Share of the permanent shock in the forecast error variance of output, h quarters ahead
- Allowing the ratios to trend (services gaining share) raises the role of the permanent shock considerably at short horizons; extending the sample changes it again; the great-ratio LR rises to 24.3 on data to 2025
- Conclusion: the size of “trend shocks” in the cycle is not pinned down by the data alone

Recap: Structural VECM

- Cointegration splits shocks into $n - r$ permanent and r transitory ones and supplies rk long-run zeros
- With one common trend the permanent shock is $\alpha'_{\perp} \varepsilon_t$, scaled
- KPSW: balanced growth gives a clean identification, but the variance shares depend on the deterministic case and the sample

From a VECM to a conditional ECM (1/2)

- Partition $y_t = (y_t, x_t')'$, x_t of dimension k ; if x_t is weakly exogenous and there is one relation involving y_t , the conditional model is

$$\Delta y_t = c_0 + c_1 t + \pi_{yy} y_{t-1} + \pi'_{yx} x_{t-1} + \sum_{i=1}^{p-1} \psi_i \Delta y_{t-i} + \sum_{j=0}^{q-1} \omega'_j \Delta x_{t-j} + u_t$$

- Notation Pesaran, Shin and Smith (2001)

- ▶ y_t : the dependent variable; x_t : the k forcing variables; c_0, c_1 : intercept and trend coefficient
- ▶ π_{yy} (scalar), π_{yx} ($k \times 1$): the coefficients on the lagged levels, i.e. the long-run part
- ▶ ψ_i, ω_j : short-run coefficients; p, q : lag orders; u_t : the error, uncorrelated with Δx_t

From a VECM to a conditional ECM (2/2)

- This is an $ARDL(p, q)$ in levels, rewritten
 - ▶ long-run coefficients $\theta = -\pi_{yx}/\pi_{yy}$: the equilibrium is $y = \theta'x$ (plus constant)
 - ▶ speed of adjustment $\pi_{yy} \in (-1, 0)$: the share of last period's disequilibrium corrected this period; half-life $\ln 0.5 / \ln(1 + \pi_{yy})$
- Inference
 - ▶ OLS is consistent and $\hat{\theta}$ has a mixed-Gaussian limit when the lags absorb the endogeneity of x_t Pesaran and Shin (1998)
 - ▶ standard errors of $\hat{\theta}$ by the delta method; t -tests on θ are asymptotically valid
- The attraction: x_t may be $I(0)$, $I(1)$ or a mix, and we need not pre-test it

The bounds test (1/2)

- H_0 : no level relationship, $\pi_{yy} = 0$ and $\pi_{yx} = 0$
 - ▶ F : the Wald statistic of these $k + 1$ restrictions, divided by $k + 1$
 - ▶ t : the t ratio of $\hat{\pi}_{yy}$; it excludes the degenerate case $\pi_{yy} = 0 \neq \pi_{yx}$
- The null distribution depends on the integration order of x_t
 - ▶ lower bound: all of x_t is $I(0)$; upper bound: all of x_t is $I(1)$
- Decision
 - ▶ F above the upper bound: reject, there is a level relationship
 - ▶ F below the lower bound: do not reject
 - ▶ between the bounds: inconclusive, the order of integration of x_t matters

The bounds test (2/2)

- ▣ Five cases, as in Johansen
 - ▶ I: no deterministic terms; II: restricted intercept
 - ▶ III: unrestricted intercept; IV: unrestricted intercept, restricted trend
 - ▶ V: unrestricted intercept and trend
 - ▶ in cases II and IV the restricted terms are part of H_0 ; the t -bounds exist for cases I, III, V
- ▣ Simulated 5% bounds, $k = 1$ ($T = 1000$, 10 000 replications)
 - ▶ F : case II [3.58; 4.14], case III [4.97; 5.71]
 - ▶ t , case III: [-2.86; -3.22]

Critical values in small samples

- ▣ PSS tabulate asymptotic bounds ($T = 1000$); with 30–80 annual observations they are too low [Narayan \(2005\)](#)
- ▣ Our simulation, case III, $k = 1$, 5%: $T = 30$: [5.52; 6.32], $T = 80$: [5.02; 5.86], $T = 250$: [4.98; 5.73], asymptotic: [4.97; 5.71]
- ▣ Response-surface regressions give critical values and approximate p -values for any T , k and lag order [Kripfganz and Schneider \(2020\)](#): use them instead of the asymptotic table
- ▣ Frequent misuses in applied work
 - ▶ the wrong case (case III bounds for a model estimated in case II); I(2) regressors, for which the bounds are invalid
 - ▶ feedback from y_t to x_t (no weak exogeneity), or more than one relation among the variables

ARDL or VECM?

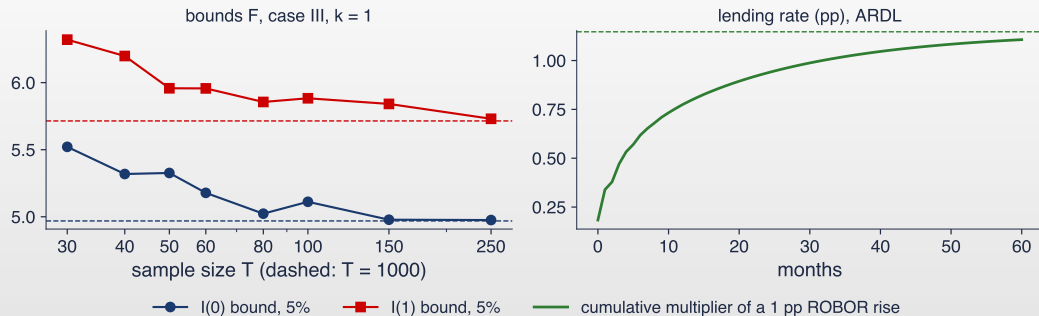
	ARDL	VECM
Relations	one, normalised on y_t	r , rank estimated
Regressors	$I(0)$, $I(1)$ or mixed; weakly exogenous	all $I(1)$; all endogenous
Inference	bounds F and t ; θ by the delta method	trace/max-eigenvalue; χ^2 tests on β , α
Small samples	few parameters; small-sample bounds needed	many parameters; bootstrap needed
Structural use	multipliers of x on y	common trends, identified shocks

- Rule: estimate the VECM first when you can
- ▶ weakly exogenous regressors and a unique relation: the ARDL is efficient and simpler
 - ▶ feedback from y_t to x_t : the ARDL estimates are inconsistent

ARDL pass-through of ROBOR to the lending rate

- ▣ ARDL(4, 2) selected by BIC, case II (ROBOR weakly exogenous in the VECM), $n = 244$
- ▣ Bounds $F = 4.27$; 5% bounds asymptotic $[3.58; 4.14]$, at our T $[3.67; 4.14]$: a level relationship
 - ▶ the same regression in case III: $F = 5.87$ against $[4.94; 5.79]$: inconclusive; $t = -3.28$ against $[-2.86; -3.22]$
- ▣ Long-run pass-through $\hat{\theta} = 1.15$ (SE 0.12): complete pass-through not rejected by the t -test; speed $\hat{\pi}_{yy} = -0.026$ (SE 0.008), half-life 27 months
- ▣ Deposit rate: $F = 7.68$, $\hat{\theta} = 0.94$ (SE 0.05), half-life 12 months: incomplete but faster

Small-sample bounds and the dynamic multipliers



- Left: simulated 5% bounds, case III, $k = 1$, against T (dashed: $T = 1000$); right: cumulative response of the lending rate to a permanent 1 pp rise of ROBOR (dashed: $\hat{\theta}$)
- Multipliers: 0.18 on impact, 0.62 after 6 months, 0.77 after 12, 0.94 after 24, 1.11 after 60

Interpreting the ARDL results

- The upper bound falls from 6.32 at $T = 30$ to 5.73 at $T = 250$: with annual data the asymptotic table over-rejects
- The verdict on a level relationship depends on the case: decide case II or III from economics (no drift in rates) before looking at F
- ARDL and VECM agree on the size of the lending pass-through (1.15 and 1.13) but not on the test of $\theta = 1$: different nuisance parameters, different power
- About half of the long-run effect arrives within six months (0.62); the rest is slow, with a half-life of the disequilibrium of 27 months, consistent with fixed-rate periods and contract repricing

Two related tools (1/2)

▣ **Toda–Yamamoto causality** Toda and Yamamoto (1995)

- ▶ a VAR of possibly integrated or cointegrated variables
- ▶ fit $p + d_{\max}$ lags in levels and test only the first p ; $d_{\max}$: the highest suspected order of integration
- ▶ the Wald statistic is asymptotically $\chi^2(p)$ whatever the integration and cointegration properties
- ▶ the price: lower power than a correctly specified VECM

▣ **Nonlinear ARDL** Shin, Yu and Greenwood-Nimmo (2014)

- ▶ partial sums of increases: $x_t^+ = \sum_{j \leq t} \max(\Delta x_j, 0)$
- ▶ partial sums of decreases: $x_t^- = \sum_{j \leq t} \min(\Delta x_j, 0)$, so that $x_t = x_0 + x_t^+ + x_t^-$

Two related tools (2/2)

□ Nonlinear ARDL: estimation and tests

- ▶ x_t^+ and x_t^- enter the ARDL as two regressors, with long-run coefficients θ^+ , θ^- and short-run coefficients ω_j^+ , ω_j^-
- ▶ long-run asymmetry: $\theta^+ \neq \theta^-$; short-run asymmetry: different ω_j^+ and ω_j^-
- ▶ both tested by Wald tests; the bounds test uses $k = 2$

□ Classic use: “rockets and feathers” [Bacon \(1991\)](#)

- ▶ retail fuel prices rise fast with oil and fall slowly
- ▶ Seminar 4 tests it for Romania

Recap: ARDL and the bounds test

- The conditional ECM is an ARDL rewritten; it needs weak exogeneity and a single relation
- The bounds test avoids pre-testing the regressors, at the price of an inconclusive zone
- Use the right case and small-sample (response-surface) critical values
- Toda–Yamamoto for causality without pre-testing; NARDL for asymmetric adjustment

Panels: the gain and the new problems



The ECB seat, Frankfurt

Photo: DXR (2015); CC BY-SA 4.0;

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- The gain: N countries add information
 - ▶ country series are short: 20–30 years of annual data
 - ▶ a single-country rank or bounds test has little power
- Asymptotics in two dimensions [Phillips and Moon \(1999\)](#), [Kao \(1999\)](#)
 - ▶ sequential ($T \rightarrow \infty$, then $N \rightarrow \infty$) or joint, with $N/T \rightarrow 0$
 - ▶ a panel regression of $I(1)$ series estimates an average long-run relation even without cointegration
- Two new problems
 - ▶ **heterogeneity**: the dynamics differ across countries
 - ▶ **cross-section dependence**: common shocks (2009, COVID-19, the euro)
 - ▶ first-generation tests assume independent units; second-generation tests model the dependence [Breitung and Pesaran \(2008\)](#)

Measuring cross-section dependence

- ▣ **CD statistic** [Pesaran \(2021\)](#): the scaled sum of all pairwise correlations

$$CD = \sqrt{\frac{2T}{N(N-1)}} \sum_{i < j} \hat{\rho}_{ij} \rightarrow N(0, 1)$$

- ▶ $\hat{\rho}_{ij}$: the correlation over time between the residuals (or demeaned series) of units i and j ; N : number of units; T : number of periods
- ▶ reading: $CD \approx 0$ under the null; $|CD| > 1.96$ rejects it at 5%
- ▣ The null is *weak dependence* [Pesaran \(2015a\)](#)
 - ▶ correlation that vanishes on average as N grows
 - ▶ a common factor with non-zero mean loadings gives $|CD| \rightarrow \infty$
 - ▶ valid for fixed T and large N , also in dynamic and unit-root panels
 - ▶ positive and negative correlations can offset each other: report the mean $|\hat{\rho}_{ij}|$ as well

Cross-section dependence in EU-27 data

- EU-27, $T = 22$ (2001–2022)
 - ▶ consumption growth: $CD = 50.3$, mean $|\hat{\rho}_{ij}| = 0.59$
 - ▶ income growth: $CD = 25.7$; inflation: $CD = 62.4$
- Interpretation
 - ▶ every statistic is far above 1.96: strong dependence everywhere
 - ▶ first-generation panel tests are not reliable for these data

Panel unit-root tests (1/2)

- ▣ **LLC** Levin, Lin and Chu (2002): one ADF regression per unit with a common ρ

$$\Delta y_{it} = \mu_i + \rho y_{i,t-1} + \sum_j \gamma_{ij} \Delta y_{i,t-j} + e_{it}$$

- ▶ y_{it} : unit i at time t ; μ_i : unit intercepts; γ_{ij} : unit-specific lag coefficients; e_{it} : errors
- ▶ H_0 : $\rho = 0$ (unit root in every unit); H_1 : $\rho < 0$, all units stationary with the same ρ
- ▶ pooled t on orthogonalised, variance-normalised residuals, adjusted by tabulated mean and variance

- ▣ **IPS** Im, Pesaran and Shin (2003): heterogeneous ρ_i ; average the individual ADF statistics t_i and standardise

$$\bar{t} = N^{-1} \sum_i t_i, \quad Z = \sqrt{N}(\bar{t} - \mathbb{E}t) / \sqrt{\text{Var}(t)} \rightarrow N(0, 1)$$

- ▶ $\mathbb{E}t$, $\text{Var}(t)$: the mean and variance of one ADF t under the null (tabulated); a very negative Z rejects
- ▶ H_1 : a non-zero fraction of units is stationary; rejection does not say which

Panel unit-root tests (2/2)

- ▣ **CIPS Pesaran (2007)**: one common factor f_t in the errors; augment each ADF with $\bar{y}_{t-1}$ and $\Delta\bar{y}_{t-j}$ (CADF), average the t -ratios
 - ▶ $\bar{y}_t = N^{-1} \sum_i y_{it}$: the cross-section average at time t ; f_t : an unobserved shock common to all units
 - ▶ the cross-section averages proxy f_t ; the distribution is non-standard and tabulated
- ▣ Under strong dependence LLC and IPS over-reject: they treat 27 correlated countries as 27 independent pieces of evidence

Panel unit roots in EU-27 data

Variable	LLC	IPS $\bar{t}$	CIPS
consumption (trend)	-12.85 (0.075)	-2.67 (0.003)	-2.38 (0.290)
income (trend)	-11.82 (0.305)	-2.09 (0.660)	-2.18 (0.566)
inflation (constant)	-10.10 (0.003)	-1.82 (0.038)	-2.36 (0.009)

- c, y : 100 ln real per capita household consumption and disposable income; π : consumption-deflator inflation; $N = 27$, $T = 22$, one lag
- p -values in brackets from 1 000 simulations of the null with independent units, for our N and T (replacing the tabulated moments)
- Consumption: IPS rejects the unit root ($p = 0.003$), CIPS does not ($p = 0.290$): the “stationarity” was a common factor; inflation is stationary by every test

Panel cointegration tests (1/2)

- ▣ **Residual-based** Pedroni (1999), Pedroni (2004): estimate $y_{it} = a_i + \delta_i t + \beta_i' x_{it} + e_{it}$ country by country and test the residuals for a unit root
 - ▶ a_i, δ_i : country intercept and trend; β_i : country cointegrating coefficients; e_{it} : the equilibrium error, $I(1)$ under H_0
 - ▶ seven statistics: four “panel” (within, common AR coefficient) and three “group” (between, heterogeneous); H_0 : no cointegration in any unit
 - ▶ they impose a common-factor restriction (short-run dynamics of x and y alike); Kao (1999) is the homogeneous special case
- ▣ **ECM-based** Westerlund (2007): test whether each country's ECM has error correction

$$\Delta y_{it} = d_i + \alpha_i y_{i,t-1} + \lambda_i' x_{i,t-1} + \sum_j a_{ij} \Delta y_{i,t-j} + \sum_j \gamma_{ij}' \Delta x_{i,t-j} + e_{it}$$

- ▶ α_i : the speed of adjustment of country i ; λ_i : its level coefficients on x ; d_i, a_{ij}, γ_{ij} : deterministic and short-run terms
- ▶ H_0 : $\alpha_i = 0$ for all i (no error correction anywhere); negative statistics reject

Panel cointegration tests (2/2)

□ Westerlund statistics

- ▶ group: $G_t = N^{-1} \sum_i \hat{\alpha}_i / \text{SE}(\hat{\alpha}_i)$, $G_a = N^{-1} \sum_i T \hat{\alpha}_i / \hat{\alpha}_i(1)$; H_1 : some countries adjust
- ▶ $\hat{\alpha}_i(1) = 1 - \sum_j \hat{a}_{ij}$: scales $\hat{\alpha}_i$ to the long-run adjustment
- ▶ panel: P_t , P_a , the same ratios for a common α ; H_1 : all countries adjust
- ▶ bootstrap critical values under cross-section dependence

□ EU-27, c on y and π (null distributions simulated)

- ▶ Pedroni group ADF (the mean of the country ADF t -ratios of $\hat{e}_{it}$): -3.05 , 5% critical value -2.73 , $p < 0.001$
- ▶ Westerlund G_t : -1.66 , 5% critical value -2.18 , $p = 0.714$

□ The two families disagree

- ▶ the residual-based test rejects, the ECM test does not
- ▶ with $T = 22$ and strong dependence, neither verdict is secure

Estimators of the long-run relation (1/3)

▣ **Panel FMOLS/DOLS** Pedroni (2001)

- ▶ FMOLS: fully modified OLS, corrected for endogeneity and serial correlation Phillips and Hansen (1990)
- ▶ DOLS: OLS with leads and lags of Δx Saikkonen (1991), Stock and Watson (1993)
- ▶ estimated per country and averaged (group mean)

▣ **Mean group (MG)** Pesaran and Smith (1995)

- ▶ estimate each country's ARDL and average the long-run coefficients: $\hat{\theta}_{MG} = N^{-1} \sum_i \hat{\theta}_i$
- ▶ consistent under full heterogeneity; $SE = s_\theta / \sqrt{N}$, s_θ : the cross-country standard deviation of $\hat{\theta}_i$
- ▶ pooled fixed effects with common slopes are inconsistent when the true slopes differ, even for large T ; for small T the Nickell bias adds Nickell (1981)

Estimators of the long-run relation (2/3)

▣ CCE Pesaran (2006)

- ▶ model: $y_{it} = \alpha_i + \beta_i' x_{it} + \gamma_i' f_t + e_{it}$
- ▶ estimation: add the cross-section averages $\bar{y}_t, \bar{x}_t$ to each regression
- ▶ f_t : unobserved common factors; γ_i : the country loadings on them; the averages $\bar{y}_t, \bar{x}_t$ absorb f_t
- ▶ valid with non-stationary factors [Kapetanios, Pesaran and Yamagata \(2011\)](#); dynamic version CS-ARDL [Chudik and Pesaran \(2015\)](#)

Estimators of the long-run relation (3/3)

- ▣ **Pooled mean group (PMG)** Pesaran, Shin and Smith (1999): a common long run inside country-specific ECMs

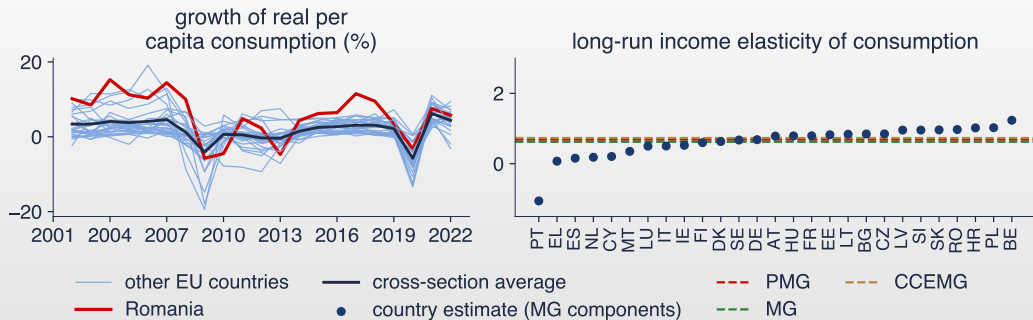
$$\Delta y_{it} = \phi_i(y_{i,t-1} - \theta' x_{i,t-1}) + \sum_j \lambda_{ij} \Delta y_{i,t-j} + \sum_j \delta'_{ij} \Delta x_{i,t-j} + \mu_i + \varepsilon_{it}$$

- ▶ θ : the long-run coefficients, common to all countries; ϕ_i : the country speed of adjustment, $\phi_i < 0$ required
- ▶ $\lambda_{ij}, \delta_{ij}, \mu_i, \sigma_i^2 = \text{Var} \varepsilon_{it}$: short run, intercept and error variance, all country-specific
- ▶ ML: maximise $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ over θ ; T_i : the sample length of country i (derivation: [Appendix](#))
- ▣ Hausman test of $\theta_{MG} = \theta_{PMG}$
 - ▶ $H = (\hat{\theta}_{MG} - \hat{\theta}_{PMG})' [\widehat{\text{Var}}(\hat{\theta}_{MG}) - \widehat{\text{Var}}(\hat{\theta}_{PMG})]^{-1} (\hat{\theta}_{MG} - \hat{\theta}_{PMG}) \rightarrow \chi^2(\dim \theta)$
 - ▶ PMG is efficient under homogeneity, MG consistent in both cases: a small H (large p-value) supports pooling

Case study: Pesaran, Shin and Smith (1999)

- Pesaran, Shin and Smith (1999)
 - ▶ consumption functions for OECD countries, annual data
 - ▶ ARDL in log real per capita consumption, log real per capita disposable income and inflation
 - ▶ they compare MG, PMG and dynamic fixed effects and use Hausman tests to decide whether the long-run income and inflation effects can be pooled
 - ▶ their argument: theory (permanent income) restricts the long run; adjustment speeds, habits and credit constraints differ across countries
- Our replication of the specification on EU-27, 2001–2022, ARDL(1,1,1) for every country, 672 observations
 - ▶ c_{it} on y_{it} and π_{it} as defined above; national currencies (country intercepts absorb the units)
- We start in 2000 to avoid the inflation episodes of the 1990s in Bulgaria and Romania, which would dominate a pooled inflation coefficient

Heterogeneity across the EU



- Left: consumption growth (Romania in red, cross-section average in dark blue); right: country long-run income elasticities from the individual ARDLs, with MG, PMG and CCEMG
- Country elasticities range from -1.06 to 1.23 ; Romania 0.97

Long-run estimates for EU-27

Estimator	θ_y	θ_π	ϕ
MG	0.63 (0.09)	0.82 (0.32)	-0.43 (0.032)
PMG	0.69 (0.04)	0.41 (0.18)	-0.31 (0.028)
dynamic FE	0.81 (0.03)	0.01 (0.23)	-0.29 (0.025)
group-mean DOLS	0.77 (0.05)	-0.24 (0.17)	—
CCEMG	0.73 (0.05)	0.10 (0.10)	—

- Standard errors in brackets (MG, DOLS, CCEMG: dispersion of country estimates; PMG: likelihood; FE: clustered by country)
- Hausman MG against PMG: $H = 2.41$, $\chi^2(2)$, $p = 0.300$; CD of the PMG residuals: 30.9

Interpreting the panel estimates

- ▣ The income elasticity lies between 0.63 (MG) and 0.81 (FE), clearly below one: consumption does not track income one for one over 2001–2022
- ▣ The Hausman test does not reject pooling, so PMG is the efficient choice for the income effect
- ▣ The inflation effect is not robust: its sign and size change with the estimator; do not report one of them alone
- ▣ PMG residuals remain strongly dependent ($CD = 30.9$): standard errors that ignore common shocks are too small; CCEMG is the safer benchmark

Panel VAR in one slide

- $y_{it} = \mu_i + \sum_{j=1}^p A_j y_{i,t-j} + \varepsilon_{it}$: common dynamics, country fixed effects [Holtz-Eakin, Newey and Rosen \(1988\)](#)
 - ▶ y_{it} : the vector of variables of country i ; A_j : lag matrices common to all countries; μ_i : country fixed effects; ε_{it} : innovations
- With small T the within estimator is biased by $O(1/T)$ (Nickell); estimate by GMM on forward-orthogonal deviations with lagged levels as instruments [Abrigo and Love \(2016\)](#)
- Identification of shocks as in [Chapter 3](#) (Cholesky, signs); impulse responses and FEVD are common to all countries
- Homogeneity of A_j is the strong assumption: with macro panels of moderate T , mean-group VARs or CCE-augmented VARs are the heterogeneous alternatives

Recap: Panel time series

- ▣ Test for cross-section dependence first; with strong dependence use CIPS and CCE, not LLC/IPS and pooled OLS
- ▣ Panel cointegration tests differ in their null and in the common-factor restriction; disagreement is informative
- ▣ MG for heterogeneity, PMG when theory pools the long run, Hausman to decide, CCE against common factors

An open question

- Is the long-run pass-through of money-market rates to bank lending rates complete in Romania, and did it change after the 2022 tightening?
 - ▶ formal: $H_0: \theta_{\text{lend}} = 1$ in a rank-2 VECM with ROBOR weakly exogenous; against a break in θ or in the markup in 2022
 - ▶ falsified by a robust rejection: across pre-registered lag lengths, deterministic cases and bootstrap inference
- Why it matters: the speed and completeness of pass-through decide how much the BNR must move its rate to move credit conditions
 - ▶ literature to start from: de Bondt (2005), Égert, Crespo-Cuaresma and Reininger (2007), Johansen (1995a), Cavaliere, Rahbek and Taylor (2012), Pesaran, Shin and Smith (2001)

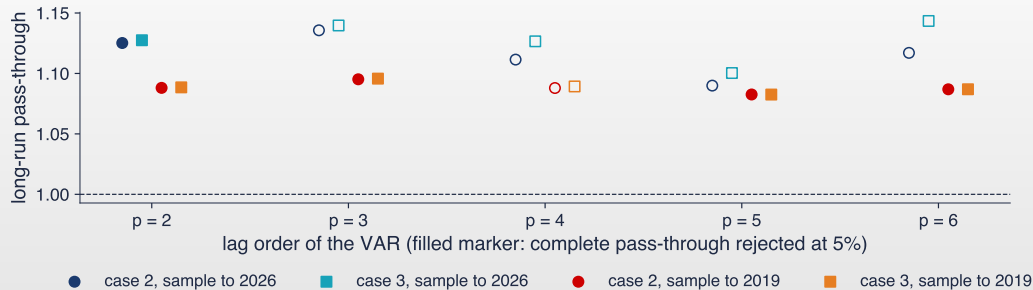
The discovery loop with an AI assistant

- An AI assistant (an LLM such as Claude, ChatGPT, Gemini or Copilot) speeds up each step; Semantic Scholar and Elicit help with the literature
 - ▶ **literature:** List peer-reviewed studies of interest-rate pass-through in Central and Eastern Europe that use cointegrated VARs or ARDL; give DOIs. Then check every DOI on Crossref
 - ▶ **hypothesis:** Propose three reasons why lending-rate pass-through could exceed one, and one testable implication of each.
 - ▶ **code and replication:** ask for a Johansen function with restricted constant, then reproduce a known number first (the eigenvalues of this lecture)
 - ▶ **robustness and critique:** Act as a hostile referee: list every specification choice that could flip the complete pass-through test.
- Report: what was asked, what was kept, what was rejected (AI_USE.md, AI_ERRORS.md)

What the human checks

- Every reference exists and says what is claimed (DOI resolves, title matches, the result is in the paper)
- The deterministic case of the critical values matches the estimated model (a frequent AI error: case-3 values for a case-2 model)
- The degrees of freedom of every LR test on β and α are counted by hand
- Lag length, case, sample and the treatment of 2008–2009 are fixed before the results; all variants are reported
- A claim of “complete pass-through” is checked against the confidence set of θ , not against one p -value

Mini-case: the estimate is robust, the verdict is not



- Long-run pass-through to the lending rate in the rank-2 VECM: lags $p = 2, \dots, 6$, cases 2 and 3, samples ending in 2019 and in 2026; filled markers: $\theta = 1$ rejected at 5%
- All 20 estimates lie between 1.08 and 1.14, yet complete pass-through is rejected in 10 of them: an AI summary that reports one p -value “with confidence” is wrong

Project idea

- **Interest-rate pass-through in Central and Eastern Europe:** replicate first, then extend
 - ▶ replicate: the Romanian rank-2 VECM of this lecture (eigenvalues 0.244, 0.177, 0.017) and the ARDL pass-through 1.15
 - ▶ extend: Romania, Hungary, Czechia and Poland; broken constants [Johansen, Mosconi and Nielsen \(2000\)](#) in 2008 and 2022; wild-bootstrap rank tests; PMG across countries with a Hausman test; NARDL for asymmetric pass-through
 - ▶ pre-register: variables, sample, case, lag rule, rank procedure, the restrictions to test and the multiple-testing correction
- Deliverables follow the course rules: repository, report, AI_USE.md, AI_ERRORS.md, oral defence

Key takeaways

- ▣ Johansen = reduced-rank regression; the rank test is non-standard, depends on the case and over-rejects in small samples
- ▣ Given the rank, restrictions on β (economics) and α (weak exogeneity) are ordinary χ^2 tests
- ▣ Cointegration supplies long-run zeros for structural analysis: permanent and transitory shocks
- ▣ ARDL is a conditional VECM: valid under weak exogeneity and a single relation, with small-sample bounds
- ▣ Panels add power but bring heterogeneity and cross-section dependence: CD first, then CIPS, MG/PMG and CCE

Self-assessment

Questions

- Why does one eigen-decomposition give the ML estimator of β for every rank?
- Which deterministic case fits two interest rates without drift?
- How many degrees of freedom has the test of a known β with $n_1 = 3$, $r = 2$?
- What does weak exogeneity of ROBOR allow you to do?
- Why can IPS reject a unit root that CIPS does not reject?

Next:

- Bayesian VAR, factor models and nowcasting
- Shrinkage handles the parameter problem we met in small samples; factors generalise the cross-section averages of CCE

Appendix: from the likelihood to the eigenvalues

- ▣ For fixed β : $|\hat{\Omega}(\beta)| = |S_{00} - S_{01}\beta(\beta'S_{11}\beta)^{-1}\beta'S_{10}|$; apply the partitioned determinant to $\begin{pmatrix} S_{00} & S_{01}\beta \\ \beta'S_{10} & \beta'S_{11}\beta \end{pmatrix}$ in both orders
- ▣ $|\hat{\Omega}(\beta)| |\beta'S_{11}\beta| = |S_{00}| |\beta'(S_{11} - S_{10}S_{00}^{-1}S_{01})\beta|$, which gives the ratio on the slide
- ▣ Write $\beta = S_{11}^{-1/2}\eta$: the ratio becomes $|\eta'(I - M)\eta|/|\eta'\eta|$ with $M = S_{11}^{-1/2}S_{10}S_{00}^{-1}S_{01}S_{11}^{-1/2}$; it is minimised by the eigenvectors of M with the r largest eigenvalues
- ▣ The minimum is $\prod_{i \leq r} (1 - \lambda_i)$; nested ranks share eigenvectors, so the LR statistics are sums of $-T \ln(1 - \hat{\lambda}_i)$

Appendix: deriving the long-run impact matrix C

- ▣ Multiply the VECM by $\alpha'_{\perp}$: $\alpha'_{\perp} \Delta y_t = \alpha'_{\perp} \sum_i \Gamma_i \Delta y_{t-i} + \alpha'_{\perp} \varepsilon_t$; summing gives $\alpha'_{\perp} \Gamma y_t \approx \alpha'_{\perp} \sum_{s \leq t} \varepsilon_s$ up to stationary terms
- ▣ Decompose $y_t = \beta(\beta' \beta)^{-1} \beta' y_t + \beta_{\perp}(\beta'_{\perp} \beta_{\perp})^{-1} \beta'_{\perp} y_t$; the first part is stationary
- ▣ Hence $\alpha'_{\perp} \Gamma \beta_{\perp} (\beta'_{\perp} \beta_{\perp})^{-1} \beta'_{\perp} y_t \approx \alpha'_{\perp} \sum_s \varepsilon_s$; invert $\alpha'_{\perp} \Gamma \beta_{\perp}$ (the $I(1)$ condition) and premultiply by $\beta_{\perp}$
- ▣ Result: the non-stationary part of y_t is $\beta_{\perp} (\alpha'_{\perp} \Gamma \beta_{\perp})^{-1} \alpha'_{\perp} \sum_s \varepsilon_s = C \sum_s \varepsilon_s$

Appendix: the PMG likelihood

$$\square \ell(\theta, \phi, \sigma^2) = - \sum_{i=1}^N \frac{T_i}{2} \ln(2\pi\sigma_i^2) - \sum_{i=1}^N \frac{1}{2\sigma_i^2} (\Delta y_i - \phi_i \xi_i(\theta))' H_i (\Delta y_i - \phi_i \xi_i(\theta))$$

□ Notation

- ▶ $\Delta y_i, y_{i,-1}$ ($T_i \times 1$), $X_{i,-1}$ ($T_i \times k$): the changes, the lagged levels of y and of x for country i
- ▶ $\xi_i(\theta) = y_{i,-1} - X_{i,-1}\theta$: the equilibrium error of country i
- ▶ $H_i = I - W_i(W_i'W_i)^{-1}W_i'$; W_i : the country's short-run regressors and intercept, which H_i projects out

- Given θ, ϕ_i and σ_i^2 are country OLS; the concentrated likelihood $-\sum_i \frac{T_i}{2} \ln \hat{\sigma}_i^2(\theta)$ is maximised over θ (back-substitution or Newton)
- $\hat{\theta}_{PMG}$ is consistent and asymptotically normal if $\phi_i < 0$ for all i and the long run is homogeneous; the $I(0)/I(1)$ status of x does not matter, as in ARDL

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